An Algorithmic Approach to Stochastic Human Discipline: Optimizing Exam Preparation via Dependency Graphs and Geometric Recovery

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Abstract. Every existing tool for exam preparation shares the same assumption: that discipline can be measured and reported back to the student, and that awareness alone will change behaviour. This assumption does not hold. This paper takes a different position: discipline is a stochastic variable to be accommodated, and exam preparation is a control problem rather than a planning problem. The proposed system closes a feedback loop around observed student behaviour through a behavioural tracker, a scheduling engine driven by a topic priority function and dependency graph, and a psychological reset condition that eliminates the backlog accumulation that causes students to abandon existing planners entirely. Computational simulation across three conditions shows that priority-directed adaptive scheduling achieves 85.7% coverage of high-priority topics against 42.9% for a static schedule, despite beginning at half the daily study hours.

Introduction

The Problem With Existing Study Tools

Students preparing for high stakes competitive examinations across the world often face the problem of lacking the discipline required to study consistently over time. Whether preparing for JEE or NEET in India, GCSE or A-Levels in the United Kingdom, the IB globally, or any other high stakes standardised examination, the pattern of failure is consistent. The tools currently available in the market to solve this problem are of three kinds, and the basic flaw in each of these tools is the same.

Existing time tracking tools measure how much time a student spends studying versus wasting time on distractions. While this data is accurate, it produces guilt without action. Students who already struggle with discipline see the data, feel bad, and either ignore it or delete the application entirely. Awareness alone does not change behaviour.

Website blockers use a master password bypass by design. In practice, this password is used with increasing frequency after the first few days. The tool assumes students have the willpower to avoid using the bypass, which is precisely the problem it was designed to solve.

Generic study planners provide a schedule from that day to the final exam date. When a student misses one day, the backlog grows. When they miss several days, the backlog becomes impossible to clear. The student abandons the plan entirely. The tool has no mechanism to recover from this failure.

All three failures share a common cause. All existing tools assume that a student discipline level is constant. In reality, discipline fluctuates daily as a function of digital distraction, fatigue, and motivational variance. No existing tool models or adapts to this reality [8].

Reframing Exam Preparation as a Control Problem

This paper proposes an alternative approach. Instead of assuming discipline and tracking deviation from it, this system models discipline as a stochastic variable and constructs a closed-loop control system around it. The behavioural tracker measures the current state of the student. The scheduling engine acts as the controller, computing the optimal next action given the state determined by the behavioural tracker. The nightly recalculation loop closes the feedback loop and determines the next state based on what the student has actually done rather than what the student should have done.

The student is not required to do anything in terms of scheduling. The system handles micromanagement entirely, including what to study, what to prioritise, for how long, and at what difficulty. Detailed mastery scores, priority rankings, and daily progress reports are surfaced not to create guilt but to give the student confidence and a clear sense of direction. Guidance replaces guilt as the motivating mechanism.

Contributions of This Paper

This paper makes the following contributions: a formal treatment of student discipline as a stochastic variable; a five-variable student state model updated daily from behavioural data; a topic priority function with dependency-graph propagation; a geometric capacity recovery model; a psychological reset condition; and a framework for empirical validation.

System Architecture

Overview

The system is divided into four layers. The first is a cross-platform behavioural tracker measuring real study and distraction times on desktop and mobile devices. The second is a conversational AI intake module that runs once at onboarding and produces a structured numeric student profile. The third is a deterministic scheduling engine generating a five-level schedule based on the student profile and dependency graph. The fourth is a nightly recalculation loop that updates all state variables from observed behaviour and regenerates the schedule for the following day.

None of these layers operate independently. The output of each layer feeds directly into the next, forming a closed loop: behaviour is measured, a schedule is generated, compliance is observed, and the schedule is updated. This process is repeated daily until the day of the exam. Artificial intelligence is used only in the second layer. All remaining components are deterministic algorithms that

produce the same output given the same input, making the system auditable and predictable.

Cross-Platform Behavioural Tracker

The behavioural tracker is a background process requiring no interaction from the student. On desktop and laptop devices, it operates as a browser extension. On mobile devices, it operates as a companion application. The tracker logs every URL accessed along with the timestamp and classifies each into one of three types: study, distraction, and neutral. YouTube is classified by channel name and video title rather than the domain alone. In cases where classification is uncertain, a popup is shown asking the student to confirm the category manually.

All data is stored locally on the student's device and is never transmitted to an external server. At the end of each day, the student is shown a summary of total productive study hours and total distraction hours.

Conversational AI Intake Module

The intake module runs once, at the beginning of onboarding. Instead of a static form, the module has a short adaptive conversation with the student. A student who reports studying two hours a day is asked different questions than one who reports studying six.

The data gathered includes the target exam, date, self-reported scores on specific topics, daily study routine, and sleep schedule. Self-reported scores are cross-referenced with short diagnostic questions to reduce the effect of students misjudging their own ability. The result is a numeric student profile, initialising all five state variables defined in Section 3.1.

Scheduling Engine

The scheduling engine is a deterministic algorithm using the numeric student profile as input and producing a five-level schedule as output. No artificial intelligence is involved at this stage. The engine applies the priority formula (Section 3.2) to rank all topics by urgency, uses the dependency graph (Section 3.3) to enforce prerequisite ordering, and uses the geometric capacity recovery curve (Section 3.5) to determine total study hours per day. The session type for each block, whether a lecture session or a practice session, is determined by the active practice ratio defined in Section 3.1.5.

Nightly Recalculation Loop

The nightly recalculation loop runs automatically every night without any input from the student. Mastery is updated first, since both the decay step and the priority recalculation depend on current mastery values. Decay is then applied to all topics not studied that day. Capacity and discipline are updated from the observed study hours, and the active practice ratio is recomputed from session classification data. Topic priorities are then recalculated and the dependency graph is re-traced to reflect any mastery thresholds that have been crossed.

Finally, the schedule for the next day is regenerated. If three or more consecutive days of non-compliance are detected, the psychological reset condition (Section 3.8) is triggered and the schedule is regenerated from the student's current position.

System Flow

The four layers operate as a continuous closed loop. The student studies during the day while the behavioural tracker records activity silently. At night, the recalculation loop updates the student state and generates the schedule for the next day. The student wakes up to an updated plan that reflects what they actually did, not what they were supposed to do. The loop never breaks, and the system requires no manual intervention at any point after onboarding.

Mathematical Framework

Student State Model

Every student is represented as a state vector at any point in time:

$$S = (M, C, K, D, A) \tag{1}$$

where $M = [m_1, \ldots, m_n]$ is the mastery score across all n topics, $C = [c_1, \ldots, c_n]$ is the confidence score, K is study capacity, D is the discipline score, and A is the active practice ratio.

Mastery

Mastery $m_i \in [0, 1]$ for each topic i is updated after each test using a gradual update rule:

$$m_i \leftarrow m_i + \alpha_t \cdot \left(\frac{s_i}{s_{max}} - m_i \right) \tag{2}$$

where s_i is the score achieved on topic i, s_{max} is the maximum possible score, and $\alpha_t = 1/n_t$ is the learning rate with n_t the number of tests taken so far. This ensures earlier tests have a greater influence on the score, which stabilises over time.

Confidence

Confidence c_i is measured as the ratio of questions attempted to total questions in a test for topic i. In environments with negative marking such as JEE, confidence is influenced by accuracy to penalise blind attempting. Confidence is tracked as a state variable but does not enter the priority function or decay model in the current formulation. Its role in scheduling is left to future work, as noted in Section 6.4.

Capacity

Capacity is updated as a smoothed average of recent observed study hours:

$$K_{t+1} = 0.8 \cdot K_t + 0.2 \cdot H_t \tag{3}$$

where H_t is the actual study hours observed on day t.

Discipline

Discipline measures the ratio of actual focused study time to scheduled study time:

$$D = \frac{\text{Act. Study Hrs}}{\text{Sched. Study Hrs}}, \quad D \in [0, 1] \tag{4}$$

It is measured by analysing the length and frequency of breaks in each study session.

Active Practice Ratio

Active Practice Ratio measures the proportion of study time spent on active problem-solving compared to passive content consumption:

$$A_t = \frac{H_t^{practice}}{H_t^{content} + H_t^{practice}}, \quad A_t \in [0, 1] \tag{5}$$

A low active practice ratio causes the scheduling engine to prioritise practice sessions over lecture sessions, independent of mastery score.

These five variables are updated daily as new behavioural data is collected, ensuring the student state always reflects the present rather than an assumption.

Topic Priority Function

Each topic i is assigned a priority score P_i:

$$P_i = W_i \times D_i \times (1 - m_i) \tag{6}$$

where W_i is exam weightage, D_i is difficulty (fixed per exam from previous year paper analysis), and m_i is the personalised mastery score. Topics with higher priority scores are scheduled first, ensuring the most important and least understood topics are always addressed first.

Dependency Graph Representation

The syllabus is represented as a directed graph $G = (V, E)$. Each arrow from topic A to topic B indicates that A must be sufficiently mastered before B is scheduled. For example, in physics: Kinematics → Newton's Laws → Work, Power, Energy → Rotational Motion.

Each edge also carries an interconnectedness score $\lambda_{AB} \in [0, 1]$ representing the conceptual strength of the dependency, used during priority propagation as described in Section 3.4.

Priority Propagation

When a student is weak in a topic, the priority of its prerequisite topics is also increased, since gaps in foundational knowledge directly affect performance in dependent topics. For topic A that is a prerequisite of topic B:

$$P_A \leftarrow P_A + \gamma \cdot \lambda_{AB} \times P_B \tag{7}$$

where $\gamma \in [0, 1]$ is a damping factor preventing unbounded accumulation at root nodes. A default value of $\gamma = 0.9$ is used. The scheduler is therefore concept-aware rather than just topic-aware.

Geometric Capacity Recovery

Let K_0 be the baseline capacity measured at intake and K_{target} be the target capacity required to complete the syllabus before the exam date. Capacity on day t is given by:

$$K(t) = \min\left(K_0 \cdot (1 + r)^t,\ K_{target}\right) \tag{8}$$

where r is the daily growth rate, adjusted based on the student's discipline score. This gradual increase allows students to build study capacity progressively rather than being assigned an unrealistic load from the start.

The exponential model is sufficient for the scope of this paper. Future work may replace this with a logistic growth function to model the natural saturation of human study capacity at a physiological maximum.

Knowledge Decay Model

Mastery of a topic decays daily according to:

$$m_{i,t+1} = 0.995 \times m_{i,t} \tag{9}$$

Topics not revisited recently will decay below the priority threshold, automatically increasing their priority and triggering rescheduling. This is in line with the forgetting curve established by Ebbinghaus [1], adapted conservatively for meaningful academic content [9].

Daily Recalculation Algorithm

Every night the system executes the following loop:

1. Update mastery scores m_i for all topics studied today
2. Apply knowledge decay to all topics not studied today
3. Update discipline score D based on today's compliance
4. Update capacity K based on today's observed study hours
5. Recompute priority scores P_i for all topics
6. Apply priority propagation through the dependency graph
7. Generate tomorrow's schedule based on updated priorities and current capacity

The daily growth rate r is adjusted based on compliance. If compliance exceeds 80%, r remains unchanged. If compliance falls below 80%, r is halved for the following day. If compliance reaches zero, r is set to zero. Upon recovery to 80%, r returns to its previous value. For sustained compliance exceeding 80% for seven consecutive days, r increases by a factor of 1.1, providing a proportional acceleration intended to scale with the student's current growth rate.

Psychological Reset Condition

If a student misses three or more consecutive days, the current schedule is deleted entirely and a fresh schedule is generated from the student's current position. The student never sees a backlog. This psychological reset trades short term coverage for long term retention of the student in the system, which is the more important outcome.

The probability of a student abandoning the plan as a function of accumulated backlog B is:

$$P_{quit}(B) = 1 - e^{-\delta B} \tag{10}$$

where B is the total accumulated backlog in hours and $\delta > 0$ is a student-specific abandonment sensitivity parameter estimated at intake from self-reported plan abandonment history. As B increases, P_{quit} approaches 1. Setting B to zero at reset returns P_{quit} to zero. For students with a high δ, the system surfaces an early warning on the first and second consecutive missed day, presenting a reduced study target rather than waiting for the full reset threshold to be reached. This model is a theoretical approximation intended to capture the qualitative relationship between backlog and abandonment probability; empirical calibration of δ is left to future work.

Simulation

This section tests the proposed system against two baseline conditions over a 180-day preparation period using a physics syllabus of 20 topics drawn from the JEE Advanced curriculum.

Condition A models a student with no formal schedule. Topics are selected randomly with a bias towards familiar topics. Test scores are drawn from a uniform distribution of 0.38 to 0.62. This student studies two hours per day on compliant days, with a decay factor of 0.997 per day.

Condition B models a student following a static schedule covering all topics sequentially at four hours per day. Topics cycle in a fixed order, with the student advancing after five sessions. Test scores are drawn from 0.45 to 0.78. Decay is 0.995 per day. No priority function, dependency graph, or capacity recovery is applied.

Condition C models a student using the proposed system. The student starts from the same two-hour baseline as Condition A. All mechanisms described in Section 3 are active.

All three conditions share the same compliance sequence, generated once with a fixed random seed. The compliance sequence assigns a full study day with probability 0.70, a partial day with probability 0.15, and a missed day with probability 0.15. Two forced absence windows are inserted at days 25–27 and days 105–107, triggering the reset condition in Condition C. By using the same compliance sequence for all conditions, the effect of the scheduling mechanism is isolated.

Single Student Recovery Journey

Figures 1 and 2 plot the same student's progress over 180 days following the proposed system. The student begins with a study capacity of two hours per day. Capacity grows geometrically toward a target of seven hours per day, following the model in Section 3.5.

Figure 1 shows actual daily study hours alongside the theoretical $K(t)$ capacity curve. After each forced absence window the reset fires and capacity recovery resumes. The student reaches approximately 4.6 hours per day by day 180, demonstrating progressive capacity growth despite two disruption events. Actual hours consistently track below the theoretical $K(t)$ curve because missed and partial days reduce the growth rate r.

Figure 2 shows the discipline score over the same period. Discipline drops to zero during each absence window and recovers following the reset. Without the reset, the student would face an unrecoverable backlog after the second absence window at day 105.

Three Condition Benchmark

Figure 3 shows weighted mastery of high-priority topics over time using a 14-day rolling average. Condition C overtakes Condition B by approximately day 60 and continues to widen the gap. The advantage is not immediate because the adaptive system spends the early period building capacity from its lower starting point. Once capacity crosses four hours, the priority function and dependency graph appear to give Condition C a compounding advantage.

The mechanism behind Condition C's coverage result is the interaction between the priority function and the dependency graph. Consider the electromagnetic chain: Electrostatics → Current Electricity → Magnetism → EMI. Since the raw priority of EMI is high, priority propagation raises the priority of every prerequisite proportionally. Electrostatics is the gate that unlocks the rest of the chain, so the scheduler concentrates there first, accumulating 143 study sessions by day 180, then proceeds down the chain. By day 180 all four topics in the chain exceed the 0.4 mastery threshold. Condition B cycles topics sequentially and does not replicate this effect.

Figure 4 shows high-priority topic coverage at day 180. Condition C covers 85.7% of high-priority topics ($W \geq 0.75$, mastery > 0.4) compared to 42.9% for Condition B and 28.6% for Condition A, despite starting with half the daily hours of Condition B, because 507 of 512 total study sessions were directed to topics with $W \geq 0.75$.

Figure 5 shows per-topic mastery at day 180 sorted by weightage. The near-baseline mastery of low-weightage topics is an emergent consequence of limited time under correct prioritisation, not a designed exclusion.

Summary of Simulation Results

Table 1. Simulation results at day 180.

Metric	A	B	C
Total study hours	268	536	879
Wtd. mastery ($W \geq 0.75$)	0.391	0.408	0.491
High-priority coverage (%)	28.6	42.9	85.7
Resets triggered	0	0	2

Condition C accumulates the most hours because geometric capacity recovery scales it from two to nearly five hours per day over 180 days. The coverage advantage of 85.7% over 42.9% for Condition B cannot be attributed to hours alone, since Condition B with 536 hours achieves only 42.9% coverage. This difference is primarily attributable to the priority function and dependency graph focusing on high weightage topics. The simulation assumes honest compliance reporting and consistent test score accuracy. Under conditions of systematic evasion or score misreporting, Condition C's advantage would narrow, as the priority function depends on the accuracy of the mastery signal it receives. Simulation code and figures are available at [10].

Related Work

Spaced Repetition and Knowledge Decay

The exponential decay of knowledge without revision was first formally described by Ebbinghaus [1], where the forgetting curve showed the rapid rate at which retention decays after initial learning. The decay model used in this paper follows this finding, conservatively adapted for meaningful academic content rather than the nonsense syllables used in the original experiments.

Building on this, Wozniak [2] developed the SM-2 algorithm, which forms the basis of spaced repetition systems such as Anki and Mnemosyne. SM-2 adjusts review

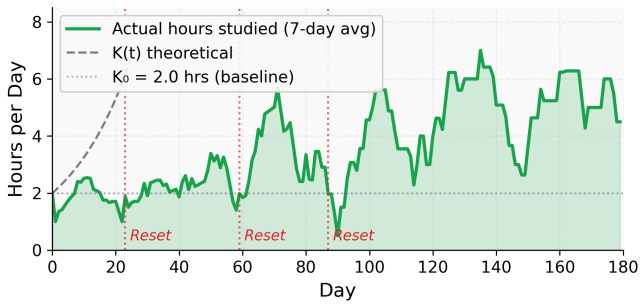

Figure 1. Geometric capacity recovery over 180 days. The student starts at 2 hrs/day and grows toward 7 hrs/day. Reset events marked in red. Despite two disruption windows the student reaches 4.6 hrs/day by day 180.

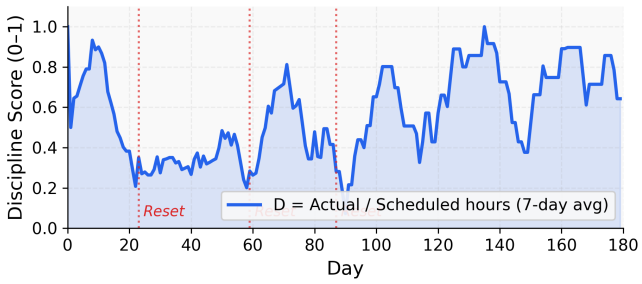

Figure 2. Discipline score over 180 days. Discipline drops to zero during each absence window and recovery restarts from a clean schedule after each reset.

timing based on recall quality, extending intervals after successful recall and resetting them after failure.

This paper differs from SM-2 in two key ways. First, mastery levels are continuously updated from test performance rather than discrete recall. Second, decay is integrated into a broader scheduling framework taking into account exam constraints, dependencies, and behavioural capacity.

Adaptive Learning Systems

Adaptive learning systems and Intelligent Tutoring Systems (ITS) aim to personalise learning by modelling the student's knowledge state. Existing research shows that such personalised learning is more effective in terms of retention and reducing knowledge decay even for voluntary users [3, 6].

Graph-based learner modelling has been found particularly effective in recent systems such as Graph-MASAL [7], which integrates a dynamic knowledge graph and multi-agent architecture to generate personalised learning paths.

This paper shares the graph-based modelling motivation but differs fundamentally in its objectives. The existing systems are content delivery systems that teach. The proposed system is a behavioural scheduling system that models student discipline as a stochastic variable and constructs a closed-loop controller around it. Content delivery is intentionally left to external platforms chosen by the student.

Stochastic Scheduling

Scheduling under stochastic conditions has been studied extensively in distributed computing, where task execution times and system load are unpredictable. Self-adapting scheduling algorithms prove that using past performance data in scheduling decisions leads to much better performance compared to traditional methods [4].

This system applies the same principle to human study behaviour. Rather than computational load, the stochastic variable is student discipline, which fluctuates daily due to distraction, fatigue, and motivation.

Closed-Loop Learning Systems

Closed-loop adaptive education systems combine learner modelling, adaptive decision-making, and feedback into a continuous loop that updates learning paths based on observed outcomes [5].

This system uses the same closed-loop structure but extends it in three ways: geometric capacity recovery, continuous knowledge decay, and a behavioural feedback signal derived from real study activity. Unlike existing systems, the control loop explicitly models both cognitive state through mastery and behavioural state through discipline and capacity.

Differentiation From Existing Work

Existing systems in spaced repetition, adaptive learning, and intelligent tutoring all assume that student engagement and discipline are either constant or externally enforced. None of the surveyed approaches model discipline itself as a dynamic state variable within the system.

The current work proposes three main innovations: modelling discipline as a stochastic process in a closed-loop system; geometric capacity recovery that scales study from the observed baseline rather than an idealised target; and a psychological reset condition that removes backlog accumulation by recreating the schedule from the current state after consecutive non-compliance.

Conclusion and Future Work

Summary of Contributions

The central argument of this paper is that existing study tools fail not because students lack information about their habits, but because no tool has modelled discipline as something that fluctuates and must be accommodated rather than assumed. The system presented here treats exam preparation as a control problem where the stochastic variable is the student's discipline, and the controller is a scheduling engine that recalculates around whatever the student actually did.

The mechanisms that make this work are not individually novel in isolation. Spaced repetition, graph-based knowledge modelling, and adaptive scheduling all have prior work. What this paper contributes is their integration into a single closed loop with a behavioural signal at its core, and specifically the geometric capacity recovery model and psychological reset condition, which together address the failure mode that breaks every existing planner: backlog accumulation after missed days.

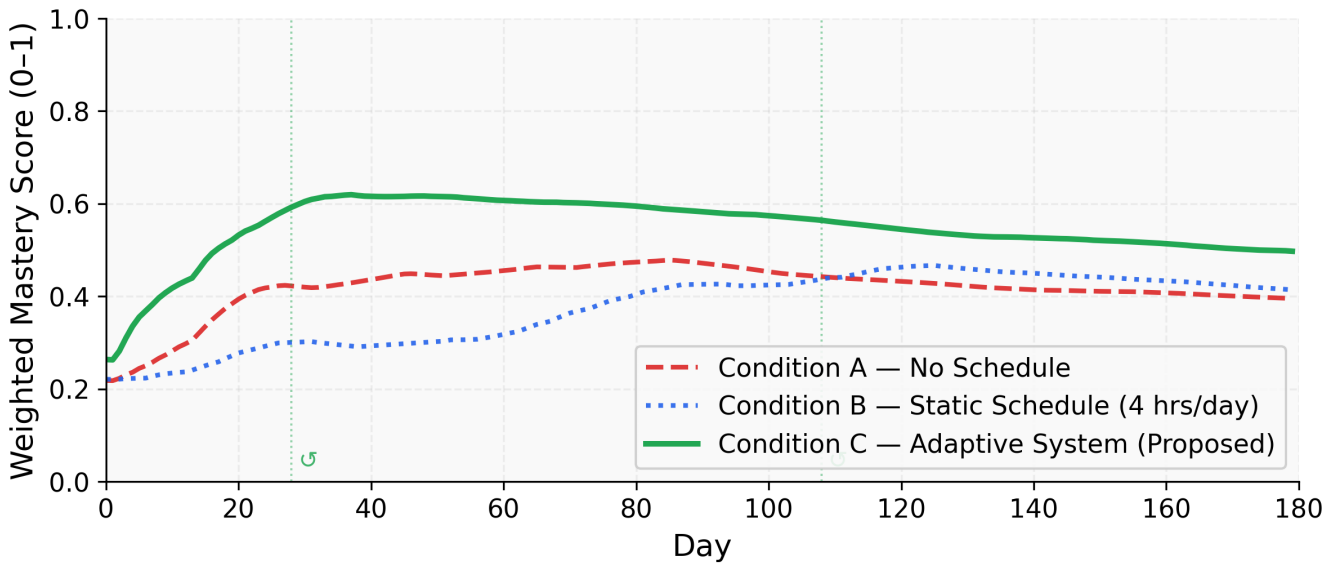

Figure 3. Weighted mastery of high-priority topics over 180 days, 14-day rolling average. Condition C overtakes Condition B around day 60 as capacity growth compounds with priority-directed scheduling.

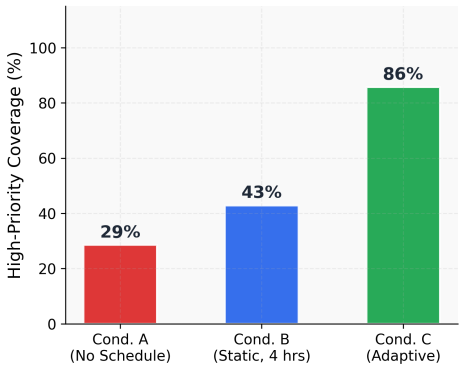

Figure 4. High-priority topic coverage at day 180. Condition C reaches 85.7% despite starting at half the daily hours of Condition B.

Limitations

The primary limitation of this work is that several key parameters remain uncalibrated from real student data. The decay constant of 0.995, capacity smoothing weights of 0.8 and 0.2, and initial state estimates at onboarding all rely on literature values and engineering convention. These will need empirical calibration once the system is deployed with real students.

A second limitation is that behavioural tracking is inherently incomplete. Standalone distraction applications, gaming consoles, and physical distractions cannot be tracked by a browser-based system. Students who deliberately evade tracking will generate inaccurate compliance data, and there is currently no mechanism to detect this.

Third, self-reported test scores cannot be independently verified. A student who consistently overreports performance will have inflated mastery scores, causing the priority function to underweight topics they have not actually mastered. Cross-validating scores against study time catches large discrepancies but not systematic overconfidence.

Empirical Validation Plan

Empirical validation is proposed through a three-condition study. Condition A establishes a behavioural baseline with no formal scheduling. Condition B measures the impact of having a static non-adaptive schedule. Condition C provides the proposed system. The metrics tracked are total productive study hours per week, syllabus coverage percentage at exam date, topic mastery scores across weak and strong areas, and student retention rate with the tool after thirty days. This design separates the effect of adaptivity from the effect of simply having any schedule at all.

Future Work

Several directions remain open. The dependency graph currently treats all prerequisite connections as uniform, and a natural extension would be weighted edge strengths to reflect varying degrees of conceptual dependence. Knowledge and confidence could also be tracked as fully independent variables, directly determining whether a lecture or practice session is scheduled on a given day. A simulation mode projecting syllabus coverage at the student's current pace would make the cost of low compliance visible in concrete terms. The Active Practice Ratio currently relies on platform classification to distinguish content consumption from problem-solving; more granular behavioural signals would improve this. The three-day reset threshold is fixed in the current model but could be personalised per student from their abandonment sensitivity parameter δ. Finally, the single-subject graph could be extended to capture cross-subject dependencies, including the reliance of Physics on Mathematics, or the overlap between Chemistry and Biology in biochemistry, allowing the scheduler to coordinate preparation across subjects simultaneously.

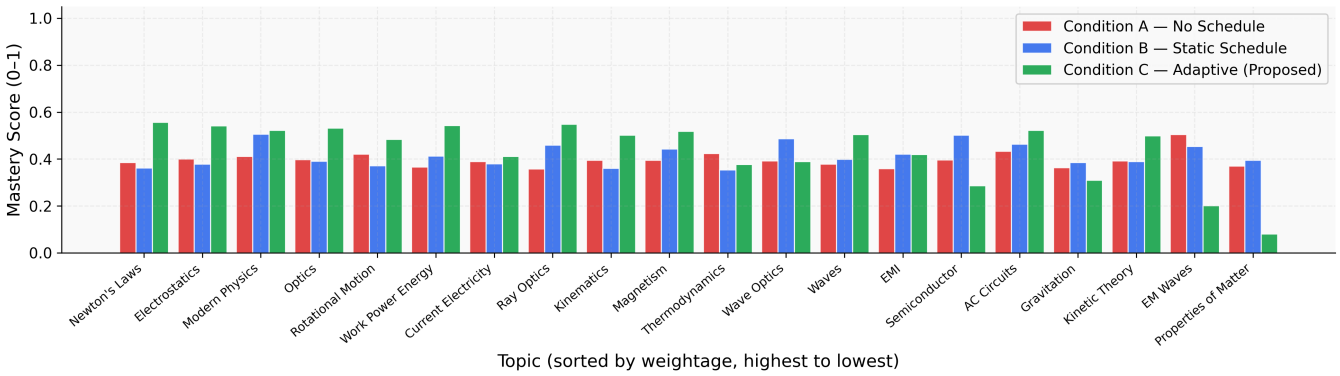

Figure 5. Per-topic mastery at day 180, sorted by weightage highest to lowest. Low-weightage topics remain near baseline as an emergent consequence of limited time under correct prioritisation.

Acknowledgments

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